

Circuitos Eléctricos II $\Rightarrow$ Minidados, pt. 3

Problema prático 11.9

$$Z = \frac{V}{I} \Rightarrow I = \frac{320 \angle 100}{72,11 \angle 33,63} = 4,43 \angle 66,31 \Rightarrow FP = \cos(\theta_v - \theta_i) = \cos(100 - 66,31) =$$

$$FP = 0,8321; S = \frac{320}{\sqrt{2}} \cdot \frac{4,43}{\sqrt{2}} = 3,1315 \cdot 229 = 709 \text{ VA}$$

Problema prático 11.10



$$Z = 10 + j4 \parallel 8 - j6 = 10 + \frac{j4 \cdot (8 - j6)}{8 + j2}$$

$$= 10 + \frac{4 \angle 90 \cdot (10 \angle 306,8638)}{8,2462 \angle 14,0362}$$

$$= 10 + \frac{40 \angle 336,8638}{8,2462 \angle 14,0362} = 10 + 4,8507 \angle 112,8336 = 10 + (-1,8823 + j4,4406)$$

$$= 8,1177 + j4,4406 = Z = 11,2673 \angle 20,044 \Rightarrow I = \frac{165 \angle 0}{11,2673 \angle 20,044} = 14,8045 \angle -20,044$$

$$FP = \cos(20,044) = 0,9336 \text{ atrasado}$$

$$P = I^2 \cdot R = 14,8045^2 \cdot 8,2673 = 2,008 \text{ kW}$$

Problema prático 11.10

$$a) \vec{S} = V_{RMS} I_{RMS}^* = (110 \angle 45) (0,4 \angle -15) = 44 \angle 70^\circ \text{ VA}$$

$$S = |\vec{S}| = 44 \text{ VA} \quad b) \vec{S} = 44 \angle 70 = 44 [\cos(70) + j \sin(70)] \approx 15,05 + j 41,35$$

$$\vec{S} = P + jQ \therefore P = 15,05 \text{ W e } Q = 41,35 \text{ VAR}$$

$$c) FP = \cos(\theta_v - \theta_i) = 0,342 = \frac{P}{S} \text{ atrasado}; \vec{Z} = \frac{V_{RMS}}{I_{RMS}} = \frac{110 \angle 45}{0,4 \angle 15} = 275 \angle 70$$

Problema práctico 11.12

$$a) \text{FP} = \cos(\theta_v - \theta_i) \Rightarrow Z = \frac{V_{\text{RMS}} \angle \theta_v - \theta_i}{I_{\text{RMS}}} = 250 \angle -75^\circ \therefore \theta_v - \theta_i = -75^\circ$$

$$F_p = \cos(-75) = 0,2588$$

$$b) Q = 10 \text{ KVAR} \Rightarrow Q = S \sin(\theta_v - \theta_i) = S \sin(-75) \Rightarrow S = \frac{10 \text{ K}}{\sin(-75)} = 10,353 \text{ KVA}$$

$$c) V_{\text{RMS}} = \sqrt{\bar{S} \cdot Z^*} \Rightarrow P = S \cos(\theta_v - \theta_i) = 10352 \cdot \cos -75 = 2679,29$$

$$\bar{S} = 2679,29 + j 10000 = 10352 \angle 75 \Rightarrow V_{\text{RMS}} = \sqrt{(10352 \angle 75)(250 \angle -75)} = 1608 \text{ V}$$